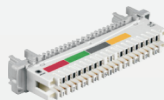


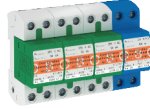
Selection aid
Overvoltage protection systems



THINK CONNECTED.

Selection aid, energy technology

AC combination arrestor and surge protection; type 1+2, type 2 and type 3

		Installation location 1 Installation in the main distributor box / combined distributor Basic protection / type 1, type 2				
Initial situation	Building type	Description	Type	Item no.	Test mark	Product figure
- No external lightning protection system - Earthing cable connection 	Private building	TN-/TT Type 2 + 3 2.5 SU Secondary counter zone	V10 Compact	5093 38 0		
			V10 Compact-AS, with acoustic signalling	5093 39 1		
	Multiple dwelling/ industry, commerce	TN-/TT Type 2 4 SU Secondary counter zone	V20-C 3+NPE	5094 65 6	VDE	
			V20-C 3+NPE+FS with remote signalling	5094 76 5	VDE	
External lightning protection system (according to DIN EN 0185-305) 	Buildings of lightning protection classes III and IV (e.g. housing, offices and commercial buildings)	TN-/TT Type 1 + 2 4 SU Secondary counter zone	V50-B 3+NPE	5093 65 4		
			V50-B 3+NPE+FS with remote signalling	5093 66 2		
Outdoor connection 	Buildings of lightning protection classes I to IV (e.g. industry)	TN-C Type 1 6 SU Pre-metered or secondary counter zone	MCD 50-B 3	5096 87 7		
		TN-S/TT Type 1 8 SU Pre-metered or secondary counter zone	MCD 50-B 3+1	5096 87 9		

Installation location 2
Installation in the subdistributor
Medium protection / type 2
Only required if distance $\geq 10\text{m}$

Description	Type	Item no.	Product figure
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TN/TT Type 2 + 3 2.5 SU	V10 Compact	5093380	
	V10 Compact-AS, with acoustic remote signalling	5093391	

TN/TT Type 2 4 SU	V20-C 3+NPE	5094656	
	V20-C 3+NPE+FS with remote signalling	5094765	

TN/TT Type 2 4 SU	V20-C 3+NPE	5094656	
	V20-C 3+NPE+FS with remote signalling	5094765	

TN/TT Type 2 4 SU	VC20-C 3+NPE	5094656	
	V20-C 3+NPE+FS with remote signalling	5094765	

Installation location 2
Installation before the terminal
Fine protection / type 3

Description	Type	Item no.	Test mark	Product figure
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Plug-in	FC-D	5092 80 0	VDE	
	FC-TV-D	5092 80 8	VDE	
	FS-SAT-D	5092 81 6	VDE	
	FC-TAE-D	5092 82 4	VDE	
	FC-ISDN-D	5092 81 2	VDE	
	FC-RJ-D	5092 82 8	VDE	
	CNS-3-D-D	5092 70 1		

Fixed installation	ÜSM-A	5092 45 1		
	ÜSM-A-2	5092 46 0		
	ÜSS 45-o-RW	6117 47 3		

Series installation in distributor	V10 Compact L1/L2/L3/N	5093 38 0		
	VF230-AC/DC	5097 65 0		
	VF230-AC-FS with remote signalling	5097 85 8		

Selection aid Photovoltaics

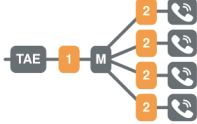
Energy technology, type 2, protection of the DC side								
Initial situation	Max. DC voltage	Max. number of MPP per inverter	Max. number of strings per MPP	Connection (DC side)	Version	Type	Item no.	Product figure
- No external lightning protection system - Earthing cable connection The following are required: - Surge voltage protection, type 2 - Lightning protection equipotential bonding 6.5 mm² 	600 V				Complete block	V20-C 3PH-600	5094 60 5	
		1	1	MC 4 connector	System solution	VG-C DCPH-Y600	5088 67 0	
	1000 V				Complete block	V20-C 3PH-1000	5094 60 8	
		1	1	MC 4 connector	System solution	VG-C DCPH-Y1000	5088 67 2	
		1	4	Terminals	PC housing	VG-C DCPH1000-4K	5088 65 0	
		1	4	Terminals	String fuse (+pole), PC housing	VG-C DCPH1000-4S	5088 65 1	
		1	6	Terminals	System solution	VG-C DCPH-MS1000	5088 69 1	
		1	6	Terminals	String fuse (+pole), PC housing	VG-C DCPH1000-6S	5088 65 2	
		2	2	MC 4 connector	PC housing	VG-C DCPH1000-21	5088 64 6	
		3	2	MC 4 connector	PC housing	VG-C DCPH1000-31	5088 64 8	

Energy technology, Type 1+2, protection of the DC side								
Initial situation	Max. DC voltage	Max. number of MPP per inverter	Max. number of strings per MPP	Connection (DC side)	Version	Type	Item no.	Product figure
- External lightning protection system according to DIN EN 0185-305 - The following are required: - Lightning and surge protection Type 1+2 - Lightning protection equipotential bonding 16 mm² - Separating distance could not be maintained 	600 V				Complete block	V50-B+C 3-PH600	5093 62 3	
		1	1		System solution	VG-BC DCPH-Y600	5088 67 6	
		1	6	Terminals	System solution	VG-BC DCPH-MS600	5088 69 3	
		1	5	Terminals	Remote signalling	VG-BC DC-MSFS600	5088 69 5	
	900 V				Complete block	V25-B+C 3-PH900	5097 44 7	
		1	1	MC 4 connector	System solution	VG-BC DCPH-Y900	5088 67 8	
		1		Terminals	System solution	VG-BC DCPH-MS900	5088 69 2	
		1	5	Terminals	Remote signalling	VG-BC DC-MSFS900	5088 69 6	
		2	2	MC 4 connector	PC housing	VG-B+C DC-DH900-21	5088 62 5	
		3	2	MC 4 connector	PC housing	VG-B+C DC-DH900-31	5088 62 9	

Data technology						
Initial situation		RJ 45	Terminal	Type	Item no.	Product figure
	- No external lightning protection system - Earthing cable connection	●		ND-CAT6A/EA	5081 80 0	
	External lightning protection system (according to DIN EN 62305)		●	FRD 24 HF	5098 57 5	

Selection aid

Telecommunications systems

Installation location 1 Installation directly on the telecommunications terminal				
Topology	Description	Type	Item no.	Product figure
Analogue telephone connection 	Basic protection in front of telecommunications system, 1 two-core wire	TKS-B	5097 97 5	
	Combined protection up to 2 two-core wires	SC-Tele 4-C-G	5081 68 8	
	Combined protection in RJ 11 version	RJ11-Tele 4-C	5081 92 0	
ISDN connection 	Basic protection in front of telecommunications system, 1 two-core wire	TKS-B	5097 97 5	
	Combined protection up to 2 two-core wires in front of NTBA	SC-Tele 4-C-G	5081 68 8	
ISDN multiplex 	Basic protection magazine up to 10 two-core wires (please order connection strip 5084008 as well)	LSA-B-MAG	5084 02 0	
	Separating strip for combination protection for 10 two-core wires	LSA-T-LEI	5084 01 2	
	Combined protection each for 1x two-core wire	LSA-BF-180	5084 02 4	
	Earthing strip for combination protection	LSA-E	5084 03 2	
	Integrated LSA solution in the housing	LSA-G	5084 04 8	
DSL and analogue telephone 	Basic protection in front of telecommunications system, 2 two-core wires	TKS-B	5097 97 5	
	Combined protection up to 2 two-core wires	SC-Tele 4-C-G	5081 68 8	
DSL + ISDN connection 	Basic protection in front of telecommunications system, 1 two-core wire	TKS-B	5097 97 5	
	Combined protection up to 2 two-core wires	SC-Tele 4-C-G	5081 68 8	

Installation location 2
Installation directly on the telecommunications terminal

Description	Type	Item no.	Product figure
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Fine protection in front of analogue terminal	RF11 Tele 4-F	5081 93 9	
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Combined protection in RJ 11 version	Combined protection in RJ 11 version	5081 92 0	
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Fine protection in front of ISDN terminal	Net Defender	5081 80 0	
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Fine protection in front of PC	Net Defender	5081 80 0	
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Alternatively as connector system	FC-ISDN-D	5092 81 2	
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Fine protection in front of ISDN terminal	Net Defender	5081 80 0	
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Fine protection in front of PC	Net Defender	5081 80 0	
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Alternatively as connector system	FC-ISDN-D	5092 81 2	
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Fine protection in front of PC	Net Defender	5081 80 0	
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Fine protection in front of analogue terminal	RJ11 Tele 4-F	5081 93 9	
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Fine protection in front of PC	Net Defender	5081 80 0	
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Fine protection in front of ISDN terminal	Net Defender	5081 80 0	
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Selection aid MSR systems

	Power supply			
Topology	Description	Type	Item no.	Product figure
Various, e.g.: - Multi-wire signal systems - Fire alarm systems	Protection of the power supply AC and DC	VF 230-AC/DC	5097 65 0	
Various sensors, e.g.: 4-20 mA - Current loops	Protection of the power supply AC and DC	VF 230-AC/DC	5097 65 0	
Bus systems/controllers	Protection of the power supply	VF 230-AC/DC	5097 65 0	
	Protection of the power supply AC with remote signalling	VF 230-AC-FS	5097 85 8	
	Protection of the power supply AC/DC with remote signalling	VF2-230-AC/DC-FS	5097 93 9	

Protection on the sensor

Description	Type	Item no.	Test mark	Product figure
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Protection of up to 10 two-core wires
(select appropriate accessories)
(please order connection strip 5084008
as well)

LSA-MAG

5084 02 0



2-wire protection for high peak currents

TKS-B

5097 97 6



4-wire protection with test function

MDP-4/D-24-T

5098 43 1

UL



4-wire protection with test function

MDP-4/D-24-T

5098 43 1

UL



2-wire protection

FLD 24

5098 60 3

UL



2-wire protection for HF applications

FRD 24 HF

5098 57 5

UL



2-wire protection for high peak currents

TKS-B

5097 97 6



4-wire protection with test function

MDP-4/D-24-T

5098 43 1

UL



Protection of RS232 applications

SD25-V24 25

5080 27 4



Selection aid MSR systems

	Power supply				
Topology	Description	Type	Item no.	Test mark	Product figure
Application with high nominal currents (wind power) 	Surge protection for 440 V/690 V network	V20-C 3+MB25+FS	5094902	VDE	
	Protection of the power supply for control units up to 230 V	VF 230-AC/DC	5097650		
	Protection for control units up to 24 V nominal voltage	VF 24 AC/DC	5097607	UL	
Sensors in Ex areas 	Protection of the power supply (non-Ex) up to 230 V		5097650		
	Protection of the power supply (non-Ex) with IR up to 230 V	VF 230-AC-FS	5097858		
	AC / DC protection of control units (not EX) to 5 V	VF 12 AC/DC	5097453	UL	
	AC / DC protection of control units (not EX) to 24 V	VF 24 AC/DC	5097607	UL	
	AC / DC protection of control units (not EX) to 48 V	VF 24 AC/DC	5097615	UL	

Protection on the sensor

Description	Type	Item no.	Test mark	Product figure
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4-wire protection with test function and nominal currents up to 10 A

MDP-4/D-24-T

5098 43 3

UL



2-wire protection without testing point

TKS-B

5097 97 6



Intrinsically safe surge protection for 3 cables in the housing (metric thread)

FDB-3-24-M

5098 38 2

Ex



Intrinsically safe surge protection for 3 cables in the housing (NPT thread)

FDB-3-24-N

5098 39 2

Ex



4-wire protection to 5 V, Ex tested

MDP-4/D-5-EX

5098 41 2

Ex



4-wire protection to 24 V, Ex tested

MDP-4/D-24-EX

5098 43 2

Ex



4-wire protection to 48 V, Ex tested

MDP-4/D-48-EX

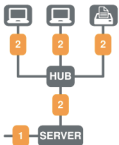
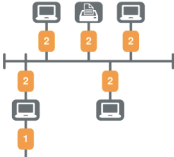
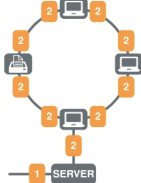
5098 45 2

Ex



Selection aid

Data technology / network technology

	Installation location 1 External communication cable			
Topology	Description	Type	Item no.	Product figure
Star topology 	Basic protection of the incoming cable (please order connection strip 5084008 as well)	LSA-B-MAG	5084 02 0	
	Combined protection of the incoming cable	SC-Tele/4-C-G	5081 68 8	
Bus topology 	Combined protection BNC connection	CoaxB-E2/MF-C	5082 41 2	
	Combined protection N connection	CoaxN-E5/MF-C	5082 46 3	
	Combined protection for two-core wire	SC-Tele/4-C-G	5081 68 8	
Various network applications 	Protection of the Wi-Fi port / Power over Ethernet	Net Defender	5081 80 0	
	IP cameras	Net Defender	5081 80 0	
	VoIP applications	Net Defender	5081 80 0	

Installation location 2 Protection on the terminal			
Description	Type	Item no.	Product figure
Data cable protection for channel CLASS EA	Net Defender	5081 80 0	
Data cable protection for channel up to Class D	RJ45-ATM/8-F	5081 79 3	
Fine protection with BNC connection (Class C)	CoaxB-E2/MF-F	5082 42 0	
Fine protection for 230 V power supply, connectable	FC-D	5092 80 0	
Fine protection for hat rail installation	VF 230-AC/DC	5097 65 0	

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Selection aid

Antennas

	Installation between BC transition point and amplifier or SAT system			
Topology	Description	Type	Item no.	Product figure
HF applications (overview) 	N connection	DS-N (m/f)	5093 99 6	 
	S-UHF connection	S-UHF (m/f)	5093 02 3	 
	BNC connection	DS-BNC (m/m)	5093 26 0	 
	TNC connection	DS-TNC (m/f)	5093 27 0	 
	7/16 connection	DS-7 16 (m/f)	5093 17 1	
	SMA connection	DS-SMA	5093 27 7	 
SAT protection 	LNB protection / receiver	DS-F m/f	5093 27 5	 
	LNB protection / receiver	DS-F f/f	5093 27 2	 
	Protection device for Multiswitch (4x SAT; 1x terrestrial)	TV 4+1	5083 40 0	
CCTV application 	Protection of IP-based CCTV	Net Defender	5081 80 0	
	Protection of CCTV camera (coaxial)	Coax B-E2 MF-F	5082 42 0	
CATV 	F connection	DS-F m/f	5093 27 5	 
		DS-F f/f	5093 27 2	 

Installation directly on the terminal				
Description	Type	Item no.	Test mark	Product figure
Fine protection for terminal	FineController FC-SAT-D	5092 81 6	VDE	
Fine protection for terminal	FineController FC-TV-D	5092 80 8	VDE	
Fine protection for terminal	FineController FC-SAT-D	5092 81 6	VDE	
Fine protection for terminal	FC-D	5092 80 0	VDE	
Fine protection for terminal	FineController FC-SAT-D	5092 81 6	VDE	

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OBO BETTERMANN GmbH & Co. KG

P.O. Box 1120
58694 Menden, Germany

Customer Service Germany

Tel. +49 (0)2373 891 500
Fax +49 (0)2373 897 777
E-mail: info@obo.de

www.obo-bettermann.com

THINK CONNECTED.